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We will discuss several questions from Chapters 2 and 3 in Ross' book.

Tutorial Questions

- (1) In an experiment, die is rolled continually until a 6 appears, at which point the experiment stops. What is the sample space of this experiment? Let E_n denote the event that n rolls are necessary to complete the experiment. What points of the sample space are contained in E_n ? What is $\left(\bigcup_{i=1}^{\infty} E_n\right)'$?
- (2) If 8 rooks (castles) are randomly placed on a chessboard, compute the probability that none of the rooks can capture any of the others. That is, compute the probability that no row or file contains more than one rook.
- (3) There are n socks, 3 of which are red, in a drawer. What is the value of n if, when 2 of the socks are chosen randomly, the probability that they are both red is $1/2$?
- (4) If $P(E) = 0.9$ and $P(F) = 0.8$, show that $P(E \cap F) \geq 0.7$. In general, prove Bonferroni's inequality, namely,

$$P(E \cap F) \geq P(E) + P(F) - 1.$$
- (5) Show that if $P(A_i) = 1$ for all $i \geq 1$ then $P\left(\bigcap_{i=1}^{\infty} A_i\right) = 1$.
- (6) Celine is undecided as to whether to take a French course or a chemistry course. She estimates that her probability of receiving an A grade would be $1/2$ in a French course and $2/3$ in a chemistry course. If Celine decides to base her decision on the flip of a fair coin, what is the probability that she gets an A in chemistry?
- (7) Three cards are randomly selected, without replacement, from an ordinary deck of 52 playing cards. Compute the conditional probability that the first card selected is a spade given that the second and third cards are spades.
- (8) A laboratory blood test is 95 percent effective in detecting a certain disease when it is, in fact, present. However, the test also yields a "false positive" result for 1 percent of the healthy persons tested. (That is, if a healthy person is tested, then, with probability .01, the test result will imply that he or she has the disease.) If 0.5 percent of the population actually has the disease, what is the probability that a person has the disease given that the test result is positive?

Assignment Questions

- (1) A , B , and C take turns flipping a coin. That is, A flips first, then B , then C , then A , and so on. The sample space of this experiment can be defined by

$$S = \left\{ H, TH, TTH, TTTH, \dots, TTTT \dots \right.$$

- (a) Interpret the sample space.
 (b) Say that the first one to get a head wins. Define X as the event that A wins, Y as the event that B wins. Find $(X \cup Y)'$.
- (2) Two symmetric dice have both had two of their sides painted red, two painted black, one painted yellow, and the other painted white. When this pair of dice is rolled, what is the probability that both dice land with the same colour face up?
- (3) Two dice are thrown n times in succession. Compute the probability that double 6 appears at least once. How large need n be to make this probability at least $1/2$?
- (4) Prove Boole's inequality:

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} P(A_i).$$

- (5) Suppose that an urn contains 8 red balls and 4 white balls. We draw 2 balls from the urn without replacement.
- (a) If we assume that at each draw each ball in the urn is equally likely to be chosen, what is the probability that both balls drawn are red?
- (b) Now suppose that the balls have different weights, with each red ball having weight r and each white ball having weight w . Suppose that the probability that a given ball in the urn is the next one selected is its weight divided by the sum of the weights of all balls currently in the urn. Now what is the probability that both balls are red?
- (6) Suppose we have 10 coins such that if the i -th coin is flipped, heads will appear with probability $i/10$, $i = 1, 2, \dots, 10$. When one of the coins is randomly selected and flipped, it shows heads. What is the conditional probability that it was the fifth coin?
- (7) An ordinary deck of 52 playing cards is randomly divided into 4 piles of 13 cards each. Compute the probability that each pile has exactly 1 ace.
- (8) In a certain species of rats, black dominates over brown. Suppose that a black rat with two black parents has a brown sibling.
- (a) What is the probability that this rat is a pure black rat (as opposed to being a hybrid with one black and one brown gene)?
- (b) Suppose that when the black rat is mated with a brown rat, all 5 of their offspring are black. Now what is the probability that the rat is a pure black rat?
- (9) A bin contains 3 different types of disposable flashlights. The probability that a type 1 flashlight will give over 100 hours of use is .7, with the corresponding probabilities for type 2 and type 3 flashlights being .4 and .3, respectively. Suppose that 20 percent of the flashlights in the bin are type 1, 30 percent are type 2, and 50 percent are type 3.
- (a) What is the probability that a randomly chosen flashlight will give more than 100 hours of use?
- (b) Given that a flashlight lasted over 100 hours, what is the conditional probability that it was a type j flashlight, $j = 1, 2, 3$?